Criterion B practice: toy-car investigation plan

STUDENT CRITIQUE ACTIVITY

Read the experimental plan below. For each section A1–A4, identify what is strong, what is unclear or missing, and how the plan could be improved. Use the MYP Sciences Criterion B rubric provided by your teacher.

A1 — Research question and explanation

How does the velocity of a toy car change as the time elapsed increases when it rolls down a ramp at a constant angle? Velocity will be determined as it travels along the ramp.

A2 — Hypothesis with scientific reasoning

I predict the car will move faster the longer it travels, because cars usually gain speed when going downhill.

A3 — Variables

Independent variable: elapsed time since release. Dependent variable: velocity of the car. Controlled variables: use the same ramp, the same car and do not change the ramp setup.

A4 — Method and materials

Materials: toy car, ramp, paper, pen, ruler and stopwatch. Mark distances along the ramp. Put the car at the top and let go. Start a stopwatch and record how long it takes to reach each mark. Estimate velocities between markers, draw a velocity–time graph and describe the trend.

Assessment practice

Section	Provisional level / judgement	What is missing or weak?	Specific improvement
A1			
A2			
A3			
A4			

Final judgement: What is the single most important improvement you would make to the plan, and why?